

Module 05: Shaping the Manifold, Stitch by Stitch — How Actions Sculpt Topology

Learning Objectives

1. Identify how increases, decreases, and stitch placement function as **topological operators** that control Gaussian curvature in crocheted fabric.
2. Recognize the structural parallel between crochet shaping decisions and **neural network error correction** (backpropagation and gradient descent).
3. Map the crocheter's action selection process — choosing stitch type, placement, and tension — onto the **active inference** framework, where action minimizes the gap between prediction and reality.

Introduction

Pick up a piece of crocheted fabric. Run your fingers across it. If it lies flat, someone controlled its curvature beautifully. If it ruffles at the edges like a coral reef, or cups into a bowl like a soup dish, something different happened — and that something is the subject of this module.

Every stitch is an action that changes the topology of the fabric. Not metaphorically — literally. When you work two single crochets into one stitch (an increase), you are adding surface area to the manifold without adding boundary. When you single crochet two together (a decrease), you are removing surface area. When you skip a stitch and chain over the gap, you are punching a hole — changing the genus of the surface. When you join a row into a round, you are performing surgery on the topology itself, turning an open strip into a closed tube.

This is action as geometric transformation. The crocheter does not merely decorate a surface — the crocheter *sculpts a manifold*. And the decisions that guide this sculpting — which stitch, where to place it, how tightly to pull — are precisely the kinds of decisions that active

inference describes: actions selected to minimize the difference between what you predicted (the pattern in your mind) and what you observe (the fabric taking shape under your hands).

In neural network terms, these small adjustments are like the weight updates of backpropagation — tiny, targeted corrections that collectively reshape the landscape of the system. Let us trace these connections carefully.

Key Concepts

1. Increases and Decreases as Curvature Operators

Mathematicians describe the curvature of a surface using a quantity called **Gaussian curvature**. A flat table has zero Gaussian curvature everywhere. A sphere has positive Gaussian curvature — it curves the same way in every direction. A saddle (or a Pringle chip, or a hyperbolic crochet coral) has negative Gaussian curvature — it curves upward in one direction and downward in the other.

Here is the remarkable thing: a crocheter controls Gaussian curvature with two simple operations.

Increases (positive stitch growth): When you work 2 sc in one stitch, you are locally increasing the surface area of the fabric without increasing its boundary length. The fabric has more material than a flat surface needs. Where does the extra go? It ruffles. It waves. It produces **negative Gaussian curvature** — a hyperbolic surface. This is why Daina Taimina's famous crocheted hyperbolic planes work: by increasing in every stitch (or every other stitch), the crocheter forces the fabric into a permanently hyperbolic geometry. The more frequently you increase, the more dramatically the surface ruffles.

Decreases (negative stitch growth): When you sc2tog (single crochet two together), you are removing surface area — pulling two columns of stitches into one. Now the fabric has less material than a flat surface needs. It cups. It bowls. It curves inward. This is **positive Gaussian curvature** — a spherical surface. The crown of a crocheted hat, the top of an amigurumi head, the curve of a crocheted bowl: all are shaped by strategic decreases that force positive curvature.

Flat fabric (zero curvature) requires a precise balance. A flat circle in single crochet needs exactly 6 increases per round (starting with 6 in the magic ring, then 12, then 18, and so on — always adding 6). Too many increases per round and it ruffles. Too few and it cups. The crocheter who achieves a perfectly flat circle has solved a curvature equation with their hands.

Now here is the connection to neural networks. In gradient descent — the algorithm that trains most neural networks — the network makes small adjustments to its weights to reduce the difference between its output and the desired output. Each weight adjustment is a tiny change to the shape of the **loss landscape**, the mathematical surface that describes how wrong the network is for every possible combination of weights.

Increases and decreases are the crocheter's gradient steps. Each one is a small, local adjustment that changes the shape of the fabric landscape. An increase in the wrong place creates a ruffle (an error); a decrease in the wrong place creates a pucker (a different error). The crocheter, like the neural network, is navigating toward a target shape through a series of small corrective operations. The flat circle's rule of "6 increases per round" is a learned gradient — a reliable step direction that the crocheter applies round after round to stay on the path toward flatness.

2. Stitch Placement as Topological Surgery

Curvature is only part of the story. The crocheter also performs genuine **topological surgery** — operations that change the fundamental connectivity of the surface.

Loop selection: A standard single crochet goes under both top loops of the stitch below, connecting to both strands. But when you work into the **front loop only (FLO)**, you leave the back loop unworked — creating a ridge on the back of the fabric and subtly changing how that row connects to the one below. Working into the **back loop only (BLO)** does the mirror operation. These are not merely decorative choices; they change the fabric's bending behavior, its drape, and the structural connectivity between rows. In topological terms, you are choosing which edges of the mesh to connect through, altering the graph structure of the fabric.

Skip stitches and chains: When you skip a stitch and chain over the gap, you are creating a hole in the surface. In topology, holes matter enormously — they define the **genus** of a surface. A flat disc has genus 0 (no holes). A torus (donut shape) has genus 1. Lace crochet,

with its regular pattern of skipped stitches and chain spaces, is a fabric of higher genus — a surface riddled with deliberate holes, each one a topological operation performed by the crocheter's hands.

Joining rounds: When you work back and forth in rows, your fabric is topologically a strip — it has two edges (top and bottom) and two sides. The moment you join the last stitch to the first with a slip stitch, you perform surgery: you have glued two edges together, turning the strip into a tube. The fabric now has only one boundary (the open end) and its topology has fundamentally changed. If you then decrease to close that open end, you have sealed the tube into a closed surface — a sphere, an egg, a blob. This is the topological journey of every amigurumi: strip to tube to closed surface, accomplished through joining and decreasing.

Working in the round with a twist: If you join a strip with a half-twist before connecting, you create a Mobius strip — a one-sided surface. Crocheters have made these, sometimes deliberately, sometimes by accident (that frustrating moment when your cowl has a twist you did not intend). The twist is a topological operation, and it cannot be undone by stretching or reshaping — only by cutting (frogging back to the join) and reconnecting.

In neural network architecture, topology matters too. The way neurons connect — feedforward layers, skip connections, recurrent loops — defines the network's computational topology. A feedforward network is like a strip: information flows in one direction. A recurrent network is like a tube: information loops back. Skip connections are like the crocheter working into a stitch several rows below, creating a shortcut in the fabric's connectivity graph. The structural decisions of the network architect parallel the topological decisions of the crocheter.

3. Action Selection in Active Inference

Now we arrive at the heart of it: how does the crocheter decide what to do?

In the active inference framework, an agent maintains a **generative model** — an internal prediction of how the world works and what it should look like. The agent then selects **actions** to minimize **free energy**: the difference between the predicted state of the world and the observed state. When prediction and reality match, free energy is low and the agent is content. When they diverge, free energy is high and the agent must act.

For the crocheter, the generative model is the pattern — the mental image of the finished object, informed by written instructions, diagrams, or past experience. The crocheter predicts: "If I work 6 increases evenly spaced in this round, the circle will stay flat." Then they crochet the round and observe: is it flat? If yes, prediction confirmed, free energy low, continue. If it ruffles, prediction error detected — too many increases, or increases placed too close together. The crocheter must now select a corrective action: fewer increases in the next round, or perhaps rip back and redistribute the increases more evenly.

This is exactly the structure of active inference: **perceive** the state of the fabric, **compare** to the prediction, **select an action** to reduce the discrepancy, **execute** the action, and **observe** the new state. The cycle repeats with every stitch, every row, every round.

Backpropagation in neural networks follows the same logic. The network produces an output, compares it to the target, computes the error, and then propagates that error backward through the layers — adjusting each weight by a small amount to reduce the error. The crocheter's error propagation is more physical but structurally identical: you see the ruffle (output error), trace it back to the round where you added too many increases (the layer where the weight was wrong), frog back to that round (backpropagate), and re-crochet with the corrected increase count (adjust the weight).

Frogging deserves special attention here. In neural network training, sometimes you realize that a whole sequence of gradient steps has gone in the wrong direction — you have been descending into a local minimum that is not the global minimum. The solution is to roll back: reset the weights to an earlier state and try a different path. Frogging is the crochet equivalent. You rip back not just one stitch but rows, rounds, entire sections — rolling back your gradient steps to a point where the fabric was still on the right track. It is costly (all that work undone), but it is the only way to escape a topological dead end.

The crocheter's **action space** — the full set of available actions at any moment — is rich and continuous. They can choose stitch type (sc, hdc, dc, tr, and dozens of specialty stitches). They can choose placement (front loop, back loop, both loops, into the space, around the post, into a stitch several rows below). They can choose tension (tight, loose, somewhere in between). They can choose to increase, decrease, or maintain the stitch count. They can choose to continue or to frog. Each of these choices is a dimension in the action space, and the crocheter

navigates this high-dimensional space guided by their generative model — the pattern, the mental image, the feel of the fabric under their fingers.

Applications

The interplay of curvature operators, topological surgery, and active inference action selection shows up vividly in several areas of crochet practice:

- **3D amigurumi shaping:** An amigurumi animal is a collection of topological operations. The body starts as a magic ring (a disc), increases expand it into a hemisphere (positive curvature), straight rounds extend it into a cylinder (zero curvature), and decreases close it into a sphere. The legs are tubes (joined rounds). The ears might be flat triangles (strategic decreases from a chain). Every three-dimensional feature is sculpted by the crocheter's deliberate curvature control — increases where the surface must expand, decreases where it must contract, and flat sections where it must hold steady.
- **Garment construction — darts, waist shaping, necklines:** A crocheted sweater is not a flat rectangle. It must accommodate the three-dimensional topology of the human body. Waist shaping uses decreases to pull the fabric inward, then increases to flare it out again at the hips. Bust darts use short rows or strategic increases to add extra surface area where the body curves outward. A neckline is a hole — a topological operation that changes the genus of the yoke from 0 (no holes, a solid panel) to 1 (one hole for the head). Each of these shaping operations is the garment designer's active inference in action: predicting the body's shape and sculpting the fabric to match.
- **Mathematical crochet sculptures:** The Crochet Coral Reef project, Daina Taimina's hyperbolic planes, Hinke Osinga and Bernd Krauskopf's crocheted Lorenz manifold — these are mathematical surfaces made real through crochet. In each case, the crocheter follows a precise rule (increase every stitch, increase every other stitch, follow a computed algorithm) that generates a specific curvature profile. The result is a physical object that you can hold in your hands and feel — the topology made tangible. These projects demonstrate that crochet is not merely *like* topology; crochet *is* a method of constructing topological surfaces.

Conclusion

Every stitch is a topological operation. Increases inject negative curvature; decreases inject positive curvature. Skipped stitches punch holes. Joined rounds close strips into tubes. The crocheter, guided by their generative model of the intended shape, selects actions to minimize the difference between prediction and reality — sculpting a manifold stitch by stitch, round by round. This is active inference made tangible, backpropagation made physical, gradient descent performed with a hook and a strand of yarn.

In the next module, we turn from action to learning: how does the crocheter acquire these shaping skills in the first place? How does the network of hands get trained?

Key Terms

Term	Definition
Gaussian curvature	A measure of how a surface curves at a point; positive for spheres, negative for saddles, zero for flat surfaces
Increase	Working two or more stitches into a single stitch of the previous row, adding surface area and tending toward negative curvature
Decrease	Combining two or more stitches into one (e.g., sc2tog), removing surface area and tending toward positive curvature
Genus	The number of holes in a surface; a disc has genus 0, a torus has genus 1, lace fabric has higher genus
Topological surgery	An operation that changes the fundamental connectivity of a surface — joining, cutting, twisting
Gradient descent	An optimization algorithm that makes small adjustments to minimize a loss function; analogous to the crocheter's incremental shaping corrections
Backpropagation	The process of propagating error backward through a neural network to adjust weights; analogous to tracing a shaping error back to the round where it originated
Frogging	Ripping back completed crochet to correct errors; the crocheter's equivalent of rolling back gradient steps
Action space	The full set of actions available to an agent at any moment; in crochet, this includes stitch type, placement, tension, and the decision to continue or frog
Free energy	In active inference, the discrepancy between the agent's predictions and observations; the crocheter acts to minimize it by shaping the fabric to match the intended form

Term	Definition
Loss landscape	The surface describing how far a neural network's output is from the target for all possible weight configurations; analogous to the space of possible fabric shapes
Manifold	A mathematical surface that may curve through higher-dimensional space; crocheted fabric is a physical manifold